Machine Learning - Graduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following can only be used when training data are linearly separable?
   (a) Linear hard-margin SVM.
   (b) Linear Logistic Regression.
   (c) Linear Soft margin SVM.
   (d) The centroid method.

2. Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs. Statement 2| The ResNet-50 model has over 1 billion parameters.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

3. Statement 1| As of 2020, some models attain greater than 98% accuracy on CIFAR-10. Statement 2| The original ResNets were not optimized with the Adam optimizer.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

4. Which of the following are the spatial clustering algorithms?
   (a) Partitioning based clustering
   (b) K-means clustering
   (c) Grid based clustering
   (d) All of the above

5. Computational complexity of Gradient descent is,
   (a) linear in D
   (b) linear in N
   (c) polynomial in D
   (d) dependent on the number of iterations

True / False (10 marks)

Answer True or False

6. True or False: Stuart Russell is a prominent figure associated with discussing existential risks posed by artificial intelligence.

7. True or False: Predicting the amount of rainfall in a region is an example of clustering in machine learning.

8. True or False: Out-of-distribution detection is commonly referred to as anomaly detection in machine learning.

9. True or False: Using sampling with replacement in bagging is the primary method that prevents overfitting in ensemble models.

10. True or False: RoBERTa pretrains on a corpus that is approximately 10 times larger than the corpus used for BERT pretraining.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the rank of the matrix A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]]. Explain how its properties relate to concepts in machine learning and why rank is significant in this context.

12. Discuss the relationship between PCA and spectral clustering in terms of their eigendecomposition processes, and evaluate the assertion that logistic regression is a special case of linear regression.

End of Paper